



Scenario:

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of

Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- **Identify** security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- **Protect** internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.
- **Detect** potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- **Respond** to contain, neutralize, and analyze security incidents; implement improvements to the security process.

Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Incident report analysis

Summary	The multimedia company faced a DoS attack which shut down operations for approximately 2 hours before service was back online. The attack was depicted by a sudden flood of ICMP packets through a misconfigured firewall, which led to the server getting overwhelmed. The security team eventually applied a new set of standards to address the attack. These new implementations included rate limits for ICMP packets, source IP address validation, network monitoring software, and an IDS/IPS system to filter out traffic.
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Identify	The attack identified was an ICMP flood attack, a form of a DoS attack that floods a system with ICMP packets to overwhelm a server and shut down operations of the network.
Protect	One area that needs to be secured the most is proper firewall configuration. The team applied a rate limit for ICMP packets, but they still need to apply the same for other areas of port filtering, for example, the site could still be vulnerable to SYN flood attacks if the team has not configured a rate limit for that as well. Other areas of improvement include port filtering for unused ports to decrease unwanted traffic, proper network monitoring (which the team implemented after the attack happened), and an IDS/IPS combination to network traffic as a whole, not just for ICMP packets. What is demonstrated is a reactive security posture, but in reality the organization needs a proactive posture against other forms of attacks outside the one that just happened.
Detect	The team implemented a network monitoring system after the attack, but there was no mention of a SIEM system being used to help filter out alerts for severity. Another thing to mention is the team implemented an IDS/IPS system after the attack, which should be linked to the SIEM if not already done. Some other tools that can be used for detection are Wireshark and/or tcpdump to see log info directly, alongside being linked to a SIEM system to help have more well rounded security monitoring.
Respond	The incident management team quickly contained the attack by blocking incoming ICMP packets and taking non-critical network services offline to prevent the server from being completely overwhelmed. The cybersecurity team investigated the security event to trace the attack vector, determining that a malicious actor exploited an unconfigured firewall. Procedures should be established to clearly communicate security event statuses internally with IT staff and externally with affected end users during an active network outage.
Recover	The team successfully restored critical network services once the malicious ICMP flood was mitigated and safety controls were put in place. Procedures should be continuously updated to ensure affected assets and system data are safely brought back online without reintroducing vulnerabilities. Clear restoration updates and communication channels must be maintained internally with IT

	personnel and externally with impacted end users to confirm when normal operations have resumed
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Reflections/Notes:

- Moving from a purely reactive response (like blocking packets *after* an attack hits) to a proactive posture (implementing continuous monitoring, rate limiting, and proper firewall configurations ahead of time) is essential for long-term network security.
- Using the NIST CSF helps bridge the gap between technical incident response and strategic risk management, ensuring that lessons learned directly improve future identification, protection, detection, response, and recovery capabilities